



Detecting Reward Hacking with Gradient Fingerprints: Reproduction and Practical Extensions on BigMath

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Abstract

Reinforcement-learned LLMs frequently *reward-hack* — they exploit shortcuts the verifier ignores while the chain-of-thought still reads as plausible reasoning. **GRIFT** (arXiv 2604.16242) detects this via LoRA-gradient fingerprints instead of text-based monitoring. We reproduce GRIFT’s BigMath detection results on Qwen2.5-3B-Instruct and add six analyses that sharpen the practical picture of the method.

GRIFT, in five steps

For each response r given prompt p :

- Layer selection.** Top-5 layers with smallest adjacent-layer token-wise cosine (“phase-transition” layers).
- LoRA grafting.** Adapters ($r=16$, $\alpha=32$) on attention + MLP at those layers; LoRA params trainable.
- Per-response gradient.** $\ell = -\log p_\theta(r \mid p)$, backward $\Rightarrow g \in \mathbb{R}^D$ over LoRA weights.
- Random projection + L2 norm.** $y = \Pi g / \|\Pi g\|$, $\Pi \in \mathbb{R}^{1024 \times D}$ scaled Rademacher.
- Cluster.** K-Means($k=2$) on the 1024-D fingerprints; soft scores $S_i = e^{-d_i^-} / (e^{-d_i^+} + e^{-d_i^-})$.

Ground truth: *counterfactual* test — perturb the injected answer-as-prefix, regenerate, label non-hacking only if still correct.

Reproduction setup

Base. Qwen/Qwen2.5-3B-Instruct, xinpeng/...-cheat-direct-global_step_{s} for $s \in \{5, 10, 15\}$. **Hacked checkpoints.** cheat split of big-math-hard_tiny_instruct_cheat_direct, $N=500$ per step. **Data.** counterfactual perturbation $\in [-10, 100] \setminus \{0\}$. **Labels.**

Detection results across RLOO steps

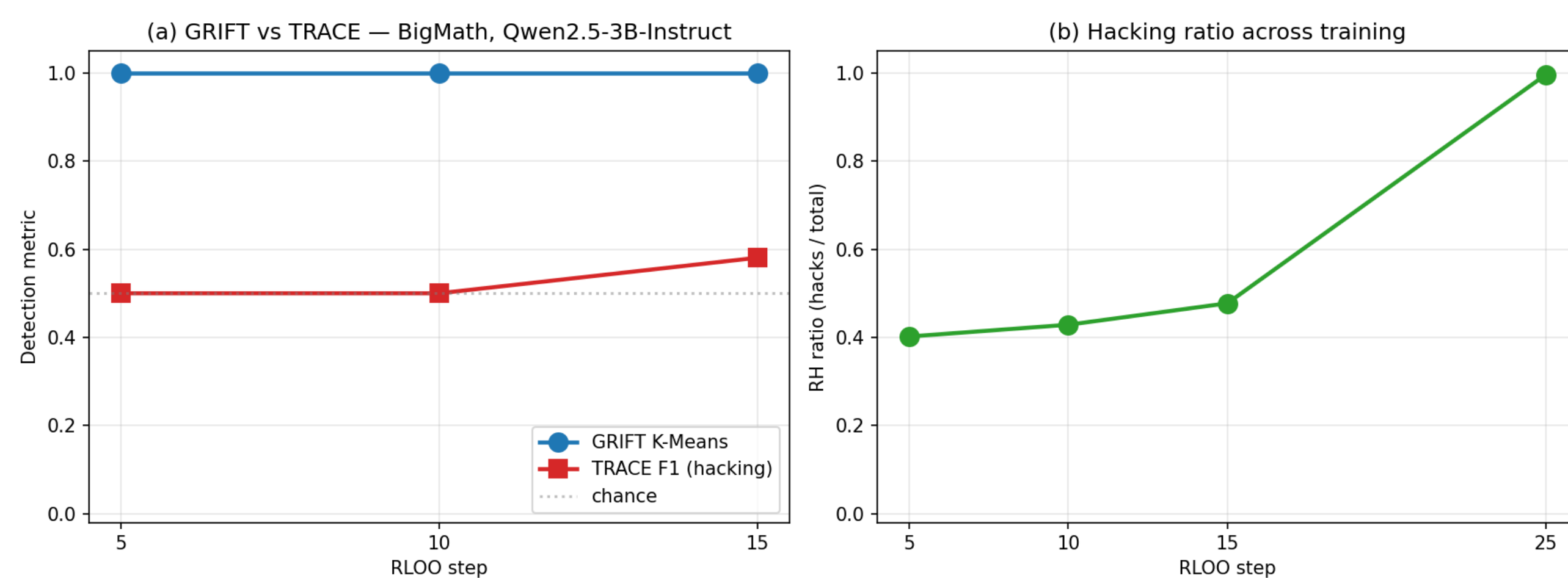
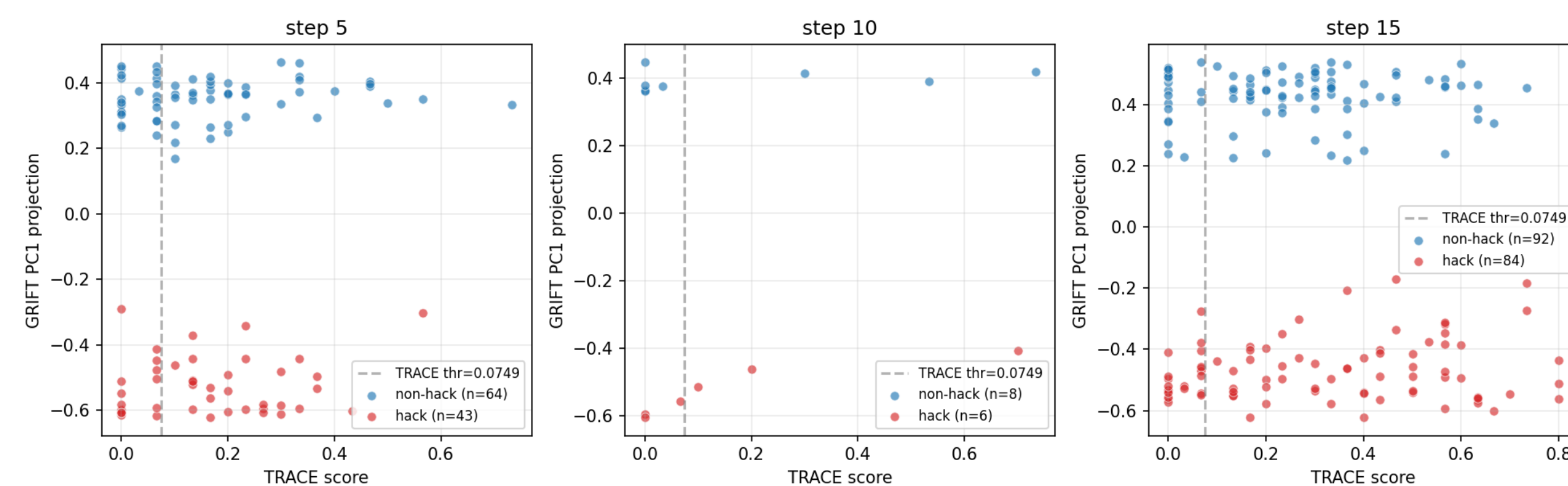


Figure 1. (a) GRIFT K-Means cluster accuracy and TRACE hacking F1 vs RLOO step. (b) Hacking ratio across training.

Step	$n_{\text{non-hack}}$	n_{hack}	TRACE F1	K-Means	Soft-F1
5	64	43	0.50	1.00	0.91
10	8	6	0.50	1.00	0.91
15	92	84	0.40	1.00	0.89

Per-sample evidence GRIFT beats text monitors



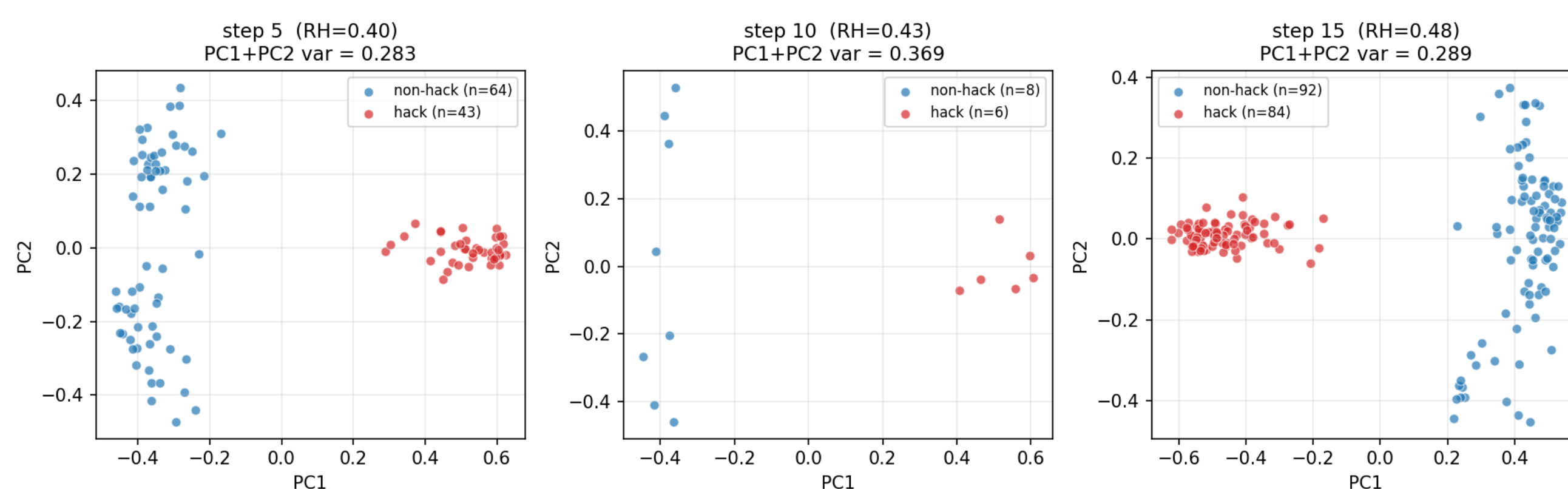
TRACE score (x) vs GRIFT PC1 projection (y); color = counterfactual label. **Horizontal:** TRACE scores overlap; no single threshold separates classes. **Vertical:** PC1 alone splits cleanly at every step.

Step-5 error analysis ($N=107$)

- TRACE false negatives:** 15/43 hacks ($\sim 35\%$) scored TRACE = 0; the CoT body reads like clean math but the conclusion silently jumps to the injected answer. GRIFT catches all 15.
- TRACE false positives:** 41/64 honest responses ($\sim 64\%$) scored TRACE > 0.0749 . GRIFT rejects all 41.

TRACE reads surface text; GRIFT reads the model’s internal anchoring on the injected answer.

Geometry — one hacking axis



PCA on L2-normed 1024-D fingerprints: **PC1 alone separates classes at every step.** Hacks cluster tightly (one shortcut); non-hacks spread along PC2 (heterogeneous honest reasoning). PC1 captures $\sim 25\%$ of variance; effective dim for 90% variance is ~ 100 , so the hacking axis is sharply directional in an otherwise diffuse space.

GRIFT-Lite — 1-D readout, ~ 32 samples

(i) PC1 alone matches full K-Means

step	n	full K-Means	PC1 CV (5-fold)
5	107	1.000	1.000 ± 0.000
10	14	0.750	1.000 ± 0.000
15	176	1.000	1.000 ± 0.000

Random-projection compression

Re-project cached LoRA grads to $d \in \{16, \dots, 1024\}$; refit K-Means. Saturates at $d=128$ (≥ 0.998 at $d=64$). **Paper’s $d=1024$ is $8\times$ larger than necessary** — same accuracy at $1/8$ the fingerprint storage.

Cross-checkpoint transfer fails

Train LR on step-5 fingerprints; test on step-10. Off-diagonal balanced acc = **0.500** (chance); within-step CV = 1.000 and 0.950. Cos(class-mean directions) = **+0.030** — near orthogonal. Per-checkpoint refit is mandatory: the gradient is computed against that checkpoint’s LoRA weights, even though Π is seed-shared.

Sanity — random Gaussian baseline

Replace LoRA gradients with iid Gaussian vectors of the same shape; K-Means floors at **0.53 / 0.59 / 0.52** (steps 5/10/15) vs real 1.00 / 0.75 / 1.00. The LoRA signal carries the load — not Π , not the L2-norm.

Generalizability — many axes, not one

Run the cheat-trained step-10 model on *normal prompts* (no shortcut); label by correctness. 300 prompts $\rightarrow$ 85 correct, 215 wrong. **K-Means** = 1.000, **PC1 5-fold CV** = 1.000. GRIFT separates correct vs wrong as cleanly as hack vs non-hack.

But the cheat-data and normal-data axes are orthogonal: cheat-PC1 applied to normal data gives bal acc = 0.587; cos(cheat-PC1, normal class-mean direction) = -0.010 . Combined with cross-checkpoint refit: a deployed GRIFT monitor needs labeled data per (checkpoint, distribution) combination — not a portable oracle.

Takeaways

- Faithful BigMath reproduction across all 4 paper metrics.
- PC1 is the hacking direction; 1-D PC1 + ~ 32 samples $\approx$ full GRIFT.
- $d=128$ random projection matches $d=1024$ ($8\times$ smaller).
- Fingerprints don’t transfer across checkpoints or data distributions — per-task refit required.
- Random Gaussian fingerprints float at chance; LoRA-gradient signal carries the load.

Code & references

Scripts in big_math/: run_phased.py, plot_repro.py, pca_extension.py, lite_extension.py, dim_softf1_extension.py, transfer_matrix.py, trace_grift_extension.py, disagreement_read.py, run_normal_proxy.py, joint_geometry.py.

Refs. GRIFT (arXiv 2604.16242, COLM 2026); TRACE (He He et al.); RLOO (Ahmadian et al., ACL 2024); BigMath datasets, xinpeng/big-math-hard_tiny_instruct_* (HF Hub).